

List

```
In [1]: # list is ordered sequence  
# we can use diff data type in list also
```

```
In [2]: l1 = []
```

```
In [4]: print(type(l1))  
  
<class 'list'>
```

```
In [5]: l1 = [10,30,60]  
l1
```

```
Out[5]: [10, 30, 60]
```

```
In [6]: list3 = [10.77,30.66,60.89]
```

```
In [7]: list4 = ['one','two' , "three"]
```

```
In [8]: list5 = ['Asif', 25 ,[50, 100],[150, 90]]
```

```
In [9]: list6 = [100, 'Asif', 17.765]
```

```
In [10]: list7 = ['Asif', 25 ,[50, 100],[150, 90] , {'John' , 'David'}]
```

```
In [11]: len(list6)
```

```
Out[11]: 3
```

```
In [13]: list3[0]
```

```
Out[13]: 10.77
```

```
In [14]: list4[0]
```

```
Out[14]: 'one'
```

```
In [15]: list4[0][0]
```

```
Out[15]: 'o'
```

```
In [16]: list4[-1]
```

```
Out[16]: 'three'
```

```
In [17]: list5[-1]
```

```
Out[17]: [150, 90]
```

```
In [18]: mylist = ['one' , 'two' , 'three' , 'four' , 'five' , 'six' , 'seven' , 'eight']
```

```
In [19]: mylist[0:3]
```

```
Out[19]: ['one', 'two', 'three']
```

```
In [20]: mylist[2:5]
```

```
Out[20]: ['three', 'four', 'five']
```

```
In [21]: mylist[:3]
```

```
Out[21]: ['one', 'two', 'three']
```

```
In [22]: mylist[:2]
```

```
Out[22]: ['one', 'two']
```

```
In [23]: mylist[-3:]
```

```
Out[23]: ['six', 'seven', 'eight']
```

```
In [24]: mylist[::-1]
```

```
Out[24]: ['eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one']
```

```
In [25]: mylist[:]
```

```
Out[25]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [26]: # add ,remove,change items
```

```
In [27]: mylist
```

```
Out[27]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [28]: mylist.append('nine')  
mylist
```

```
Out[28]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [30]: mylist.insert(9,'ten')  
mylist
```

```
Out[30]: ['one',  
          'two',  
          'three',  
          'four',  
          'five',  
          'six',  
          'seven',  
          'eight',  
          'nine',  
          'ten',  
          'ten']
```

```
In [31]: mylist.remove('ten')
```

```
In [32]: mylist
```

```
Out[32]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [33]: mylist.pop()
```

```
Out[33]: 'ten'
```

```
In [34]: mylist
```

```
Out[34]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [35]: mylist.pop(2)
```

```
Out[35]: 'three'
```

```
In [36]: mylist
```

```
Out[36]: ['one', 'two', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [37]: del mylist[7]
```

```
In [38]: mylist
```

```
Out[38]: ['one', 'two', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [39]: mylist[0]=1  
mylist[2]=4  
mylist
```

```
Out[39]: [1, 'two', 4, 'five', 'six', 'seven', 'eight']
```

```
In [40]: mylist.clear()
```

```
In [41]: mylist = ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [43]: mylist1 = mylist.copy()
```

```
In [44]: id(mylist1),id(mylist)
```

```
Out[44]: (1578915397824, 1578914701312)
```

```
In [46]: list1 = ['one', 'two', 'three', 'four']  
list2 = ['five', 'six', 'seven', 'eight']
```

```
In [47]: list1.extend(list2)  
list1
```

```
Out[47]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [48]: list1
```

```
Out[48]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [49]: list1.reverse()
```

```
In [50]: list1
```

```
Out[50]: ['eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one']
```

```
In [51]: list1 = list1[::-1]
```

```
In [52]: list1
```

```
Out[52]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [53]: mylist3 = [9,5,2,99,12,88,34]
mylist3.sort() # Sort list in ascending order
mylist3
```

```
Out[53]: [2, 5, 9, 12, 34, 88, 99]
```

```
In [54]: mylist3 = [9,5,2,99,12,88,34]
mylist3.sort(reverse=True) # Sort list in ascending order
mylist3
```

```
Out[54]: [99, 88, 34, 12, 9, 5, 2]
```

```
In [55]: L1 = [1,2,3,4,0]
```

```
In [56]: all(L1)
```

```
Out[56]: False
```

```
In [57]: any(L1)
```

```
Out[57]: True
```

```
In [58]: L2 = [1,2,3,4,True,False]
```

```
In [60]: all(L2)
```

```
Out[60]: False
```

```
In [61]: any(L2)
```

```
Out[61]: True
```

```
In [ ]:
```